

The Chinese University of Hong Kong, Shenzhen

Course Outline

2025-26 Term 2

1. Course identity

A. Course as listed at CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	STA5012
Course title (English)	Statistical Learning
Course title (Chinese)	统计学习
Units	3
Description (English)	This course provides a comprehensive overview to statistical learning — the modern interface between statistics and machine learning. It covers both discriminative and generative modeling, emphasizing conceptual understanding, methodological development, and practical applications of modern learning techniques. Core principles such as the bias–variance tradeoff and model complexity control are highlighted throughout. Practical data analysis and demonstrations will be conducted using Python, enabling students to connect theoretical concepts with real-world implementation. By the end of the course, students will develop a solid understanding of the theoretical foundations and practical tools for data modeling and prediction.
Description (Chinese)	本课程系统介绍统计学习——统计学与机器学习之间的现代交叉领域。课程内容涵盖判别模型与生成模型，强调对核心概念的理解、方法论的构建以及现代学习技术在实际中的应用。课程将贯穿讲解偏差 – 方差权衡、模型复杂度控制等基本原理。实践环节将使用 Python 进行数据分析与示例演示，帮助学生将理论概念与现实问题中的具体实现相结合。通过本课程的学习，学生将建立对数据建模与预测相关理论基础和实践工具的扎实理解。

B. Corresponding course at CUHK

Please give details of the closest corresponding course at CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course at CUHK, please make multiple copies of the block below.

Course code	
Course title (English)	
Course title (Chinese)	
Units	
Description (English)	
Description (Chinese)	

2. **Course requisites**

Please state prerequisites and co-requisites, in terms of courses at CUHK-Shenzhen or any other requirements.

A. **Prerequisites**

CSC1001, MAT2040, STA3001

B. **Co-requisites**

C. **Exclusion**

D. **Others**

3. **Learning outcomes**

By the end of this course, students are expected to:

- understand the fundamental concepts of statistical learning, including the bias–variance tradeoff, overfitting, and model complexity control;
- describe and explain key discriminative and generative learning methods, including modern classification, self-supervised representation learning, density estimation and generative modeling techniques;
- derive and analyze core results and algorithms underlying statistical learning models;
- apply appropriate learning methods to real data, implement them using software such as Python, and justify model selection and regularization choices;
- interpret and communicate data analysis outcomes clearly, linking theoretical understanding to practical implications.

4. **Course syllabus**

Please highlight fundamental concepts involved in each topic to help students better understand what is covered in the course.

1. Introduction

- Overview of statistical learning and its connection to modern AI
- discriminative and generative learning frameworks
- Core principles: sample complexity, generalization, overfitting, and the bias–variance tradeoff

2. Discriminative analysis

- Overview of modern classification methods: from perceptron and AI classifiers
- Classification robustness and large-margin classifiers
- Representation learning: from clustering to self-supervised learning and pre-training

3. Generative Modeling

- Density estimation
- Introduction to modern generative AI (e.g., VAE, GAN, diffusion models, GPT)

4. Practical Component

- Data analysis and implementation using Python
- Applications illustrating key ideas in regression, classification, and generation

5. Assessment scheme

Component/ method	% weight
Homework	40
group project	30
Exam	30

Remarks:

6. Grade descriptor

The grade descriptor should align with specific course learning outcomes, not a generic description applicable to all courses.

Type: **General**

Grade	Description
A	Demonstrate thorough mastery at an advanced level of extensive knowledge and skills required for attaining all the course learning outcomes. Show strong analytical and critical abilities and logical thinking, with evidence of original thought, and ability to apply knowledge to a wide range of complex, familiar and unfamiliar situations. Apply highly effective programming, organizational and presentational skills.
A-	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to familiar and standard situations in a manner that is logical and comprehensive. Has the ability to express the knowledge or application with clarity.

B	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.
C	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way.
D	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials Has the ability to state the knowledge or application in simple terms.
F	For example: Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

7. Feedback for evaluation

Please specify forms of evaluation to generate feedback on classes from students.

- Formal Course and Teaching Evaluation
- Informal feedback and/or teaching assistant
- School retreat or program review

8. Readings

Recommended:

1. An introduction to statistical learning with applications in python
2. The elements of statistical learning

Others:

9. Course components

Activity	Hours/week
Lecture	3

10. Indicative teaching plan

Week	Content/ topic/ activity
1	Introduction to statistical learning; supervised vs. unsupervised learning; discriminative vs generative modeling; examples from modern AI
2	Generalization, overfitting, model complexity, and the bias–variance tradeoff
3	Discriminative analysis: overview of modern classification methods: from perceptron and AI classifiers
4	Representation learning: clustering, dimension reduction and data visualization methods
5	Representation learning: self-supervised representation learning
6	Representation learning: contrastive learning and pre-training
7	Practice in discriminative analysis: integration of theory and applications in case studies
8	Classical generative modeling: introduction to density estimation, histogram and kernel density estimation
9	Modern generative modeling: latent variable model and variational autoencoders (VAE)
10	Modern generative modeling: implicit generative model and generative adversarial networks (GAN)
11	Modern generative modeling: score-based generative models and diffusion probabilistic model (DPM)
12	Modern generative modeling: autoregressive modeling and generative pre-trained transformer (GPT)
13	Practice in generative modeling: integration of theory and applications in case studies
14	Review and project presentations

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections. Example: 150 students for lecture (x 2); 30 students for tutorials (x 10).

12. Academic honesty and plagiarism

A course outline may include subject-specific requirements on plagiarism, in addition to the University established policies.

Students must complete all coursework and assessments independently, unless AI tool use is explicitly permitted. Unauthorized or improper use of AI may constitute academic dishonesty and harm learning. Misuse includes submitting AI-generated work as one's own, using AI to cheat, or employing AI unethically. Always obtain permission before using AI in academic work, and be sure to acknowledge any AI use in submissions.

13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences? Have the details (as in this document) been approved at School or other level in CUHK- Shenzhen?

提交申请

14. Any other information**15. Version date**

Version number	1
As of (date)	2025-11-03